

Monocular Depth Estimation

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Real-Time 3D Reconstruction and 6-DoF Tracking with an Event Camera

This model tries to calculate the pose (position and orientation), the log gradient and the depth map of the image simultaneously using 3 Extended Kalman Filter (EKF).

When calculating the predictions of each EKF, the other two predictions are assumed to be ground truth.

6-DOF camera position

The current state of the camera in the world frame is saved as a 6x1 vector. The prediction of the new pose is achieved by adding the product of the estimated velocity (due to short timeframes acceleration is ignored) to the current pose. Next the actual Log Intensity Gradient is compared to the actual one across the image. Based on that the uncertainty of the model and its Kalman Gain is updated. Finally the estimate of the current camera position is updated according to the last pose, its accuracy and the Kalman Gain.

Log Intensity Gradient

The Log Intensity Gradient represents the pixel-wise change of intensity in x and y. The EKF prediction of the change is expected to be 0 due to the assumption of the intrinsic texture of the detected surfaces being constant within the small timesteps between frames. The measurement is collected by the sensor. According to the differences between the predicted gradient and the measured one the Kalman Gain is calculated. Based on that the current hypothesis on the gradient map as well as the matching covariance is determined.

Depth Map

The inverse depth map works on the principle that objects closer to the camera move faster resulting in a proportional higher rate of events. The inverse of depth is used to simplify the linearization and to give pixels without a measurement or close to infinity the value 0.

The prediction is, similar to the log intensity gradient, assumed to be the previous value without any change. The measurement is again taken from the current frame and compared to the prediction. The resulting depth map and Kalman Gain/ Covariance are saved for later iterations.

Unsupervised Event-based Learning of Optical Flow, Depth, and Egomotion

In comparison this model uses machine learning in its core to extract the same parameters. Motion blur minimization is used to control the unsupervised encoder-decoder network. The network consists of two separate models, one used for optical flow estimation, the other for egomotion and depth estimation.

While the final model works on a monocular camera, the training requires a stereo setup for the stereo loss function.

Input

Since neural networks require a fixed amount of channels, a discretized event volume is used. With this technique the entire event stream is split over a fixed amount of bins. Each event is put into bins close by proportional to their time difference. Thanks to this a fixed amount of channels is achieved while retaining the full spatiotemporal distribution of the events.

Optical Flow Prediction

This model tries to acquire the pixel wise motion vector (in x and y). The loss functions used to train this model are:

- Time Loss: targets to eliminate motion blur (similar to first algorithm) by trying to rectify all images to the average timestamp of the entire datastream so that the sharpness is maximised
- Optical Flow Loss: tries to maximize the smoothness of optical flow between pixels by penalizing the spatial gradient of the flow field

Egomotion and Depth Prediction

Aims to predict the egomotion (6DOF movement of the camera) as well as depth of each pixel in the image. Loss functions:

- Temporal Projection Loss: similar to Time Loss this functions tries to minimize motion blur in the resulting image
- Stereo Disparity Loss: tries to match the image from the left to the image from the right camera to resolve metric scale for the resulting image
- Left-Right Disparity Consistency Loss: this loss enforces that the depth predicted from the left camera view is geometrically consistent with the depth predicted from the right camera view

- Smoothness Regularization: penalizes the spatial gradient of the predicted disparity map to ensure the depth result is **piecewise smooth**. It is weighted by the event image's intensity gradient